

Dynamic Cutoffs Beyond the Sieve Range: A Closed Large-Divisor Strip and the Remaining Affine Möbius Parity Fibers

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Abstract

The matched divisor expansion of a primitive fixed-shift Möbius residual has coefficient $b_R(u) = -\mu(u) \log u$ beyond its finite Ramanujan cutoff $R = X^{1/2-\delta+o(1)}$. Earlier work closes only the selected packet $u, v \leq R$. We replace the static support by a dynamic cutoff $S = X^{s+o(1)} > R$, without changing the finite model, and prove that the conductor-rectangularized tensor and induced-Gauss estimates remain valid for $u, v \leq S$. If $Q = X^{\beta+o(1)}$, $J = X^{1-\beta+o(1)}$, and the inherited no-wrap and polynomial Fourier-tail hypotheses hold, the enlarged packet has a fixed-power saving whenever $s < 1 - \beta$ and

$$\eta_S = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 4s}{2}, \frac{3 - \beta - 6s}{4} \right\} > 0.$$

Consequently every strict interior point of the earlier hybrid closure region admits a genuine fixed-power strip $S = RX^\rho$ with $\rho > 0$. This closes parts of both mixed support quadrants and also the square $(R, S]^2$; it is the first estimate in this chain for divisor coefficients strictly beyond the sieve cutoff. When $3/5 < \beta < 2/3$, the hybrid argument reaches every fixed-power cutoff $S < JX^{-\eta}$.

For the remaining terms we prove an exact complementary-divisor normal form. On a row $n = mj + h$, the ultra-long tail is

$$\sum_{\substack{u|n \\ u>S}} -\mu(u) \log u = \sum_{\substack{r|n \\ r<n/S}} -\mu(n/r) \log(n/r).$$

For two rows, fixing complementary divisors r, s parameterizes the orbit by one residue class and produces two affine forms whose determinant is exactly $h(m_1 - m_2)/(r, s) \neq 0$. The unresolved coefficient is therefore an affine two-point Möbius correlation, not an arbitrary large-modulus fiber. Tao's proved logarithmically averaged two-point Elliott theorem cancels every fixed complement fiber, and hence every fixed finite complement truncation. We also prove that the absolute value of the unweighted $r = s = 1$ fiber for each fixed row pair has mass $\gg J \log^2 J$, so a coefficientwise triangle inequality on that fiber cannot close it. The missing input is quantitative uniformity for polynomially growing rows and complement fibers in a sharp dyadic orbit window. No full residual-dispersion theorem, Hardy–Littlewood asymptotic, twin-prime lower bound, or general breach of sieve parity is claimed.

Keywords: primitive Möbius tail; dynamic divisor cutoff; exact conductors; tensor large sieve; complementary divisors; affine Möbius correlations; logarithmic Elliott theorem; sieve parity.

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1 Introduction

The primitive residual isolated in TPC-18–20 has an exact centered divisor expansion on every finite target horizon. If

$$a(u) = -\mu(u) \log u,$$

then its common four-channel coefficient is

$$b_R(u) = a(u) - \lambda'_R(u), \quad \lambda'_R(u) = 0 \quad (u > R). \quad (1)$$

Thus $b_R(u) = a(u)$ beyond the finite Ramanujan cutoff. TPC-21 removes the deterministic Ramanujan mean of the selected packet $u, v \leq R$, and TPC-22 closes its centered discrepancy in explicit parameter regions by exact-conductor rectangularization [3–7].

The support remaining after that result has four quadrants:

	$v \leq R$	$v > R$
$u \leq R$	SS	SL
$u > R$	LS	LL.

(2)

Only SS was previously closed. In particular, both-large terms exist: one may take $u = m_1 j + h$ and $v = m_2 j + h$. We therefore avoid the ambiguous phrase “one-sided large-divisor terms” for the whole complement. We also distinguish support legs from the *single-indicator Fourier channels* completed in TPC-20; the latter theorem did not estimate SL or LS in (2).

1.1 A dynamic cutoff

Our first result moves the closed support itself. Choose

$$R < S \leq X^{1-o(1)}, \quad \vartheta_{R,S}(u) = b_R(u) \mathbf{1}_{u \leq S}. \quad (3)$$

This does not replace the finite model cutoff R : it truncates the *residual* divisor coefficient at a later point. The enlarged packet $u, v \leq S$ contains SS, both mixed collars, and the nonempty both-large square $(R, S]^2$.

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad S = X^{s+o(1)}.$$

The TPC-22 character coordinates, exact-conductor large sieve, and all-frequency induced-Gauss estimate remain valid with S in place of R . The actual off-diagonal row-pair mask is essential: it gives

$$\sum_{g \leq S} \sum_{m,n} |c_{m,n}^{(g)}|^2 \ll Q^2 X^\varepsilon$$

because $g \mid m - n$, not because S is below the row length. No new hypothesis $L > 2S$ is required for the hybrid argument. What changes is the denominator range $q \leq S^2$, the conductor range $F_i \leq S$, and the two-scale completion condition $HS \ll JX^{-\eta}$.

The resulting hybrid saving is

$$\eta_S(\beta, s) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 4s}{2}, \frac{3 - \beta - 6s}{4} \right\}. \quad (4)$$

There is also an occupancy route when $2s < 1 - \beta$, provided the stronger source margin $L > 2S$ is imposed. Consequently, putting

$$s = \frac{1}{2} - \delta + \rho$$

gives a genuine strip beyond R . In the hybrid region it is enough that

$$0 < \rho < \min \left\{ \frac{1}{2} + \delta - \beta, \frac{\beta + 4\delta - 1}{4}, \frac{6\delta - \beta}{6} \right\}, \quad (5)$$

while the occupancy region permits

$$0 < \rho < \frac{2\delta - \beta}{2}. \quad (6)$$

Every bound is strict and carries a fixed margin. These formulas show that every strict interior point of the TPC-22 hybrid closure region has a positive collar. The occupancy branch has the same conclusion only when its additional source margin $L > 2S$ is available. In either case, this is a rigorous estimate for genuinely new large-divisor coefficients under the inherited packet interfaces stated below.

1.2 What the next support looks like

Let $n = mj + h$. The part beyond S , after a uniformly negligible scalar centering tail is removed, is

$$\mathcal{T}_{m,S}(j) = \sum_{\substack{u|n \\ u>S}} a(u) = \sum_{\substack{r|n \\ r<n/S}} a(n/r). \quad (7)$$

The second equality is exact and valid without assuming that n is squarefree. Since $\mu(k) = \lambda(k)\mu^2(k)$, it also gives the Liouville-cocycle identity

$$\mathcal{T}_{m,S}(j) = -\lambda(n) \sum_{\substack{r|n \\ r<n/S}} \lambda(r)\mu^2(n/r) \log(n/r). \quad (8)$$

Thus divisor reflection turns geometric size into a target-dependent parity phase.

More usefully for two rows, fix complement divisors r, s . Their two divisibility conditions select either no orbit points or one class $j = j_0 + [r, s]t$. The quotient divisors are affine forms

$$U(t) = \frac{m_1(j_0 + [r, s]t) + h}{r}, \quad V(t) = \frac{m_2(j_0 + [r, s]t) + h}{s},$$

and

$$\det(U, V) = \frac{h(m_1 - m_2)}{(r, s)} \neq 0. \quad (9)$$

The remaining LL coefficient is therefore $\mu(U(t))\mu(V(t)) \log U(t) \log V(t)$. This exact normal form identifies the next analytic input: a quantitatively uniform two-point affine Möbius estimate for growing coefficients, summed over moving complement fibers.

For each fixed fiber, the logarithmically averaged estimate is already known: Tao's two-point Elliott theorem applies because (9) is nonzero [2]. We record the corresponding log-weighted corollary and a finite-complement reassembly. We also prove that the absolute value of the $r = s = 1$ fiber has positive squarefree density and mass of order $J \log^2 J$. Hence a coefficientwise absolute-value estimate cannot close that component.

The organization is as follows. [Section 2](#) fixes the exact primitive packet. [Section 3](#) extends the Ramanujan-mean bound and removes the scalar ultra-long tail. [Sections 4](#) and [5](#) prove the dynamic-cutoff closure and its exponent ledgers. [Section 6](#) derives the complementary affine forms. [Section 7](#) gives the fixed-fiber logarithmic theorem and the absolute-value no-go. [Section 8](#) states the remaining uniform gate and strict scope. The exact certificate and conclusion are in [section 9](#).

2 Primitive packet and the dynamic residual support

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and choose

$$R = \lfloor X^{1/2-\delta} \rfloor, \quad 0 < \delta < \frac{1}{2}. \quad (10)$$

The finite Ramanujan model and residual are

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad r_R(n) = \Lambda(n) - \Lambda_R(n). \quad (11)$$

In one separated primitive row packet,

$$Q = LD, \quad QJ \asymp X, \quad L > \max\{2R, 2|h|\}, \quad (12)$$

and the rows are

$$m = \ell d \asymp Q, \quad \ell \in [L/2, 2L] \text{ prime}, \quad d \in [D, 2D], \quad d \leq R^{1/2}. \quad (13)$$

We impose $(m, H) = 1$, $(j, H) = 1$, and put

$$n_m(j) = mj + h. \quad (14)$$

Then

$$(n_m(j), mH) = 1. \quad (15)$$

In particular, every divisor of a target is coprime to the appropriate row and to H ; this exact fact replaces any unnecessary comparison between that divisor and the source prime ℓ .

Let $U \asymp X$ be a finite horizon containing all positive targets in the separated packet. Define

$$\lambda'_R(u) = u \sum_{b \leq R/u} \frac{\mu(ub)\mu(b)}{\varphi(ub)} \quad (u \leq R), \quad \lambda'_R(u) = 0 \quad (u > R), \quad (16)$$

$$a(u) = -\mu(u) \log u, \quad b_R(u) = a(u) - \lambda'_R(u). \quad (17)$$

Both coefficients are zero off squarefree integers, and for every fixed $\varepsilon > 0$, uniformly for $u \leq U$,

$$|b_R(u)| \ll_\varepsilon X^\varepsilon. \quad (18)$$

Indeed, for $u > R$ this is the elementary logarithmic bound, while for $u \leq R$ it follows from the positive divisor formula for λ'_R .

TPC-20 proves the pointwise matched representation

$$Z_m(j) = \varepsilon_U(m) + \sum_{\substack{u \leq U \\ (u, mH)=1}} b_R(u) \left(\mathbf{1}_{u|n_m(j)} - \frac{1}{u} \right), \quad (19)$$

where, for every $A > 0$,

$$\varepsilon_U(m) \ll_{A,h} X^\varepsilon (\log X)^{-A}. \quad (20)$$

No prime number theorem in a growing row progression is used in this calibration.

Choose a dynamic residual cutoff

$$R < S \leq U, \quad S = X^{s+o(1)}, \quad (21)$$

and put

$$\vartheta_{R,S}(u) = b_R(u) \mathbf{1}_{u \leq S}, \quad (22)$$

$$Z_m^{\leq S}(j) = \sum_{\substack{u \leq S \\ (u, mH)=1}} b_R(u) \left(\mathbf{1}_{u|n_m(j)} - \frac{1}{u} \right), \quad (23)$$

$$Z_m^{> S}(j) = \sum_{\substack{S < u \leq U \\ (u, mH)=1}} a(u) \left(\mathbf{1}_{u|n_m(j)} - \frac{1}{u} \right). \quad (24)$$

Then $Z_m = \varepsilon_U(m) + Z_m^{\leq S} + Z_m^{> S}$ exactly. Notice that S truncates the residual expansion only; the model (11) still ends at R .

The separated row weights x_m, y_n satisfy

$$\|x\|_\infty + \|y\|_\infty \ll_\varepsilon X^\varepsilon, \quad \sum_m |x_m|^2 + \sum_n |y_n|^2 \ll QX^\varepsilon. \quad (25)$$

The actual generic Schur mask $\mathbf{m}(m, n)$ is independent of all divisor variables, obeys

$$|\mathbf{m}(m, n)| \leq 1, \quad \mathbf{m}(m, m) = 0, \quad (26)$$

and may include the distinct-source, determinant-separation, and row-gcd restrictions of TPC-18. The dynamic theorem below applies to this row-pair-only mask. A modulus-dependent mask would require a new argument.

For squarefree $u, v \leq S$, write

$$g = (u, v), \quad u = ga, \quad v = gb, \quad q = [u, v] = gab. \quad (27)$$

The compatible joint divisibility class exists precisely when $g \mid m - n$, in addition to the primitive coprimality conditions. To make the dynamic scalar object explicit, put $\mathcal{U}_q = (\mathbb{Z}/q\mathbb{Z})^\times$ and, for $\xi \in \mathcal{U}_q$, define

$$Z_{u,v}(\xi) = \sum_{m,n} x_m y_n \mathbf{m}(m, n) \mathbf{1}_{m \equiv -h\bar{\xi} \pmod{u}} \mathbf{1}_{n \equiv -h\bar{\xi} \pmod{v}}, \quad (28)$$

$$Z_{q,S}(\xi) = \sum_{\substack{u,v \leq S \\ \text{lcm}(u,v)=q}} \vartheta_{R,S}(u) \vartheta_{R,S}(v) Z_{u,v}(\xi). \quad (29)$$

Inadmissible row-modulus combinations contribute zero. Let

$$\bar{Z}_{u,v} = \frac{1}{\varphi(q)} \sum_{\xi \in \mathcal{U}_q} Z_{u,v}(\xi), \quad D_{u,v}(\xi) = Z_{u,v}(\xi) - \bar{Z}_{u,v}. \quad (30)$$

Then

$$\bar{Z}_{q,S} = \frac{1}{\varphi(q)} \sum_{\xi \in \mathcal{U}_q} Z_{q,S}(\xi), \quad D_{q,S}(\xi) = Z_{q,S}(\xi) - \bar{Z}_{q,S}. \quad (31)$$

Thus $D_{q,S}$ is a centered function on $\mathcal{U}_q$; the centering is linear in the divisor-pair decomposition, and explicitly

$$D_{q,S} = \sum_{\substack{u,v \leq S \\ \text{lcm}(u,v)=q}} \vartheta_{R,S}(u) \vartheta_{R,S}(v) D_{u,v}. \quad (32)$$

Use

$$e(t) = e^{2\pi it}, \quad e_q(t) = e(t/q), \quad c_H(r) = \sum_{\zeta \in (\mathbb{Z}/H\mathbb{Z})^\times} e_H(r\zeta).$$

For $F \in C_c^\infty(\mathbb{R})$, write $\hat{F}(t) = \int_{\mathbb{R}} F(x) e(-tx) dx$, and define

$$\mathcal{L}_{q,S} = \frac{J}{Hq} \sum_{r \neq 0} c_H(r) \hat{F}\left(\frac{rJ}{Hq}\right) \sum_{\xi \in \mathcal{U}_q} Z_{q,S}(\xi) e_q(r\bar{H}\xi). \quad (33)$$

Here $\bar{H}$ is the inverse of $H \bmod q$, which exists by target primitivity. We write $\mathcal{L}_{q,S}^{\text{mean}}$ and $\mathcal{L}_{q,S}^{\text{disc}}$ for the same linear functional with $Z_{q,S}$ replaced by $\bar{Z}_{q,S}$ and $D_{q,S}$, respectively. Hence

$$\mathcal{L}_{q,S} = \mathcal{L}_{q,S}^{\text{mean}} + \mathcal{L}_{q,S}^{\text{disc}}.$$

We retain the TPC-20 hypotheses

$$J \geq 4HX^{\varepsilon_0} \quad (34)$$

for some fixed $\varepsilon_0 > 0$, together with its polynomial bound for the total absolute Fourier-tail mass. These assumptions are inherited interfaces, not consequences of the dynamic cutoff.

Finally, TPC-20's two-scale completion applies simultaneously to all $u, v \leq S$ under the strict condition

$$HS \leq JX^{-\eta_0} \quad (35)$$

for some fixed $\eta_0 > 0$. It removes the two single-indicator and constant nonzero spectra, leaving the compatible joint packet. We refer to (35) as the *Poisson-short condition*; it must not be confused with a restriction to one support quadrant.

3 Dynamic Ramanujan mean and scalar-tail removal

Two scalar issues must be settled before the conductor argument. The principal character of the enlarged packet has to remain harmless, and the centered coefficient beyond S has to be replaced by its actual divisor sum without a formal deletion.

3.1 The mean packet for $u, v \leq S$

Let $\mathcal{Z}_1 = \sum_{m,n} |z_{m,n}|$ be the absolute row-pair mass in one separated packet. The application has

$$\mathcal{Z}_1 \ll Q^2 X^\varepsilon. \quad (36)$$

For each compatible pair $u, v \leq S$, let $\mathfrak{R}_{u,v}$ denote its row-pair mass after the primitive and row-pair masks. If $q = [u, v]$, the unit-fiber mean is

$$\bar{Z}_{q,S} = \frac{1}{\varphi(q)} \sum_{\substack{u,v \leq S \\ \text{lcm}(u,v)=q}} \vartheta_{R,S}(u) \vartheta_{R,S}(v) \mathfrak{R}_{u,v}. \quad (37)$$

This is exact and uses no row equidistribution.

For $M \geq 1$, define

$$\Delta_M(F; J) = \frac{J}{M} \sum_{r \neq 0} c_M(r) \hat{F}\left(\frac{rJ}{M}\right). \quad (38)$$

Poisson summation over reduced classes gives

$$\Delta_M(F; J) = \sum_{(n,M)=1} F(n/J) - \frac{\varphi(M)}{M} J \hat{F}(0), \quad |\Delta_M(F; J)| \ll_F \tau(M). \quad (39)$$

The mean scalar packet is exactly

$$\mathcal{L}_{q,S}^{\text{mean}} = \bar{Z}_{q,S} \Delta_{Hq}(F; J). \quad (40)$$

Lemma 3.1 (Dynamic lcm cost). *For $S \leq X^{O(1)}$ and every $\varepsilon > 0$,*

$$\boxed{\sum_{u,v \leq S} \frac{|\vartheta_{R,S}(u) \vartheta_{R,S}(v)| \tau(H[u, v])}{\varphi([u, v])} \ll_{\varepsilon, H} X^\varepsilon.} \quad (41)$$

Proof. By (18), each coefficient is $O(X^{\varepsilon/4})$ after shrinking the auxiliary epsilon. Also, uniformly for $q \leq S^2 \leq X^{O(1)}$,

$$\frac{\tau(Hq)}{\varphi(q)} \ll_{\varepsilon, H} \frac{X^{\varepsilon/4}}{q}.$$

Writing $u = ga, v = gb$, then dropping $(a, b) = 1$, gives

$$\sum_{u, v \leq S} \frac{1}{[u, v]} \leq \sum_{g \leq S} \frac{1}{g} \left(\sum_{a \leq S/g} \frac{1}{a} \right)^2 \ll \log^3(2S). \quad (42)$$

Absorb the logarithms and coefficient factors into X^ε . $\square$

Theorem 3.2 (Dynamic mean closure). *For any collection of denominators from the selected packet $u, v \leq S$,*

$$\boxed{\sum_q |\mathcal{L}_{q, S}^{\text{mean}}| \ll_{\varepsilon, F, H} Z_1 X^\varepsilon.} \quad (43)$$

Under (36), this is $O(Q^2 X^\varepsilon)$, and therefore has a fixed-power saving relative to $XQ = Q^2 J$ whenever J has a fixed positive power of X .

Proof. Insert (37) into (40), apply (39), and bound the masked compatible row mass absolutely by Z_1 . Interchanging the row and divisor sums leaves exactly (41). $\square$

This proof is a genuine extension of the TPC-21 mean theorem: that theorem stated only the supports $u \leq R$. The extension works because $b_R(u) = a(u)$ for $u > R$, not because the principal block is discarded.

3.2 Uniform removal of the ultra-long scalar tail

Put

$$A_T(m) = \sum_{\substack{u \leq T \\ (u, mH)=1}} \frac{a(u)}{u}, \quad \rho_{P, H}(m) = \frac{mH}{\varphi(mH)}. \quad (44)$$

The uniform Möbius calibration theorem of TPC-20, based only on the classical zero-free region for $\zeta(s)$, says that for every fixed $A, C > 0$, uniformly for $mH \leq T^C$,

$$A_T(m) = \rho_{P, H}(m) + O_{A, C} \left(\rho_{P, H}(m) (\log T)^{-A} \right). \quad (45)$$

In the present fixed-power ranges, one fixed C covers both $T = S$ and $T = U \asymp X$.

Proposition 3.3 (The centered ultra-long tail is its divisor tail). *Define*

$$\tau_{S, U}(m) = \sum_{\substack{S < u \leq U \\ (u, mH)=1}} \frac{a(u)}{u}. \quad (46)$$

Then, for every $A > 0$,

$$\tau_{S, U}(m) \ll_{A, h} \rho_{P, H}(m) (\log X)^{-A} \ll_{A, h, \varepsilon} X^\varepsilon (\log X)^{-A}. \quad (47)$$

Consequently

$$Z_m^{>S}(j) = \mathcal{T}_{m, S}(j) - \tau_{S, U}(m), \quad \mathcal{T}_{m, S}(j) = \sum_{\substack{u | n_m(j) \\ u > S}} a(u). \quad (48)$$

Replacing every $Z_m^{>S}$ by $\mathcal{T}_{m, S}$ in a separated two-row packet changes it by $O_B(XQ(\log X)^{-B})$ for every prescribed $B > 0$.

Proof. Equation (46) is $A_U(m) - A_S(m)$, so (47) follows by applying (45) at both endpoints. Divisors of $n_m(j)$ are automatically coprime to mH , which proves (48).

For the packet error, the elementary bounds

$$|Z_m^{\leq S}(j)| + |\mathcal{T}_{m,S}(j)| \ll_\varepsilon X^\varepsilon \quad (49)$$

follow from the divisor bound, (18), and $\sum_{u \leq U} |b_R(u)|/u \ll (\log X)^2$. Expand the product after inserting (48), use $O(Q^2 J)$ row-orbit tuples and $Q^2 J \asymp XQ$, then choose A in terms of B . $\square$

Remark 3.4 (Calibration is not cancellation of the parity core). The proposition removes only the deterministic densities $a(u)/u$. It does not estimate the signed divisor tail $\mathcal{T}_{m,S}$. In particular, its $u = n_m(j)$ term remains present whenever the target is squarefree.

4 Conductor rectangularization at a dynamic cutoff

We now prove that the centered part of the selected packet extends from $u, v \leq R$ to $u, v \leq S$. Every step is performed before summing different lcm decompositions; there is no global character rectangle shared by all divisor pairs.

4.1 Pairwise coordinates and compatibility energy

For one squarefree pair, use (27). Since g, a, b are pairwise coprime,

$$\mathfrak{X}(q) \simeq \mathfrak{X}(u) \times \mathfrak{X}(b), \quad \chi \mapsto (\alpha, \gamma), \quad (50)$$

and

$$\text{cond } \chi = (\text{cond } \alpha)(\text{cond } \gamma). \quad (51)$$

The global principal character is exactly the pair of coordinate principals.

Put

$$c_{m,n}^{(g)} = x_m y_n \mathbf{m}(m, n) \mathbf{1}_{m \equiv n \pmod{g}}. \quad (52)$$

Character orthogonality on the shared factor gives the exact pairwise transform

$$\hat{Z}_{u,v}^\times(\chi) = \overline{\chi(-h)} \sum_{m,n} c_{m,n}^{(g)} \alpha(m) \gamma(n). \quad (53)$$

There is no residual $1/\varphi(g)$, and the character on the second row is not conjugated. This is the TPC-22 rectangularization, whose proof is purely finite-group algebra and is unchanged by $u, v > R$.

Proposition 4.1 (Compatibility energy is cutoff-independent). *Under (25)–(26),*

$$\boxed{\sum_{g \leq S} \sum_{m,n} |c_{m,n}^{(g)}|^2 \ll_\varepsilon Q^2 X^\varepsilon.} \quad (54)$$

No comparison $S < L/2$ is required.

Proof. The mask removes $m = n$. Hence

$$\sum_{g \leq S} \sum_{m,n} |c_{m,n}^{(g)}|^2 \leq \sum_{m \neq n} |x_m|^2 |y_n|^2 \sum_{g|m-n} 1.$$

The last multiplicity is at most $\tau(|m - n|) \ll_\varepsilon X^\varepsilon$, independently of S . Apply (25). $\square$

The row-pair-only nature of the mask is used here and below. If the mask changed with u, v, q , the same matrix would no longer be tested over the ambient character moduli, and the tensor step would not follow.

4.2 The two estimates for one conductor cell

We recall the weighted exact-conductor large sieve in the normalization needed here. If $w \asymp W$ runs over squarefree moduli, a_n is supported on an interval of length N , and exact conductors lie in a dyadic block $f \asymp F$, then

$$\sum_{w \asymp W} \frac{1}{\varphi(w)} \sum_{\substack{\chi \in \mathfrak{X}(w) \\ \text{cond } \chi \asymp F}} \left| \sum_n a_n \chi(n) \right|^2 \ll_\varepsilon X^\varepsilon \left(F + \frac{N}{F} \right) \sum_n |a_n|^2. \quad (55)$$

The block $F = 1$ contains conductor one. The proof writes $w = fd$, applies the primitive multiplicative large sieve at the true conductor f , and sums the inducing factors d ; it is uniform in the ambient modulus W . This is Theorem 4.1 of TPC-22 [7], derived there from the primitive multiplicative large sieve [1, Chapter 7].

Put

$$\mathcal{K}_Q(F) = F + \frac{Q}{F}. \quad (56)$$

For $q = gab \asymp Y$, $ga, gb \leq S$, let $\mathcal{E}_{Y,S}(F_1, F_2)$ be the sum of the squared transforms (53), weighted by

$$\frac{|\vartheta_{R,S}(ga)\vartheta_{R,S}(gb)|^2}{\varphi(gab)}, \quad (57)$$

over all pairwise-coprime g, a, b , coordinate characters with $\text{cond } \alpha \asymp F_1$, $\text{cond } \gamma \asymp F_2$, and all inherited admissibility conditions. This is a sum of pair energies, not the norm of a fictitious common projection.

Theorem 4.2 (Dynamic tensor conductor energy). *Uniformly for $Y \leq S^2$ and $1 \leq F_i \leq S$,*

$$\boxed{\mathcal{E}_{Y,S}(F_1, F_2) \ll_\varepsilon Q^2 X^\varepsilon \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (58)$$

Proof. Fix g , enlarge the nonnegative sum by dropping divisibility and coprimality restrictions among the ambient moduli, and apply (55) in the m -coordinate. Its Hilbert-valued form permits the entire n -coordinate to remain as a vector. Apply the same theorem in the n -coordinate. This gives

$$X^\varepsilon \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2) \sum_{m,n} |c_{m,n}^{(g)}|^2.$$

Sum in g , use [proposition 4.1](#), and absorb the dyadic and divisor-pair multiplicities into X^ε . The soft coefficient bound (18) is the only property of $\vartheta_{R,S}$ used. $\square$

Let $\mathcal{L}_{u,v}^{F_1, F_2}$ denote the scalar functional (33) formed from the (F_1, F_2) -coordinate-character projection of the unweighted centered pair $D_{u,v}$; the dynamic divisor coefficients are inserted externally below. Define the pair-separated cell sum

$$\mathfrak{L}_{Y,S}(F_1, F_2) = \sum_{\substack{g, a, b \text{ pairwise coprime} \\ ga, gb \leq S, \text{ } gab \asymp Y}} |\vartheta_{R,S}(ga)\vartheta_{R,S}(gb)| |\mathcal{L}_{ga,gb}^{F_1, F_2}|. \quad (59)$$

The inherited restricted-frequency norm route, licensed by (34) and the polynomial tail-mass assumption, gives

$$\mathfrak{L}_{Y,S}(F_1, F_2) \ll_\varepsilon Q X^\varepsilon \sqrt{JY \mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (60)$$

There is a second estimate. If a character modulo the squarefree q has exact conductor $f \mid q$, its additive transform at every nonzero frequency factors into a primitive Gauss sum of size

$\sqrt{f}$ and a Ramanujan sum modulo q/f . Summing that Ramanujan factor against the smooth Fourier weight gives

$$\sum_{r \neq 0} |c_d(r)| \left| \widehat{F} \left(\frac{rJ}{Hq} \right) \right| \ll_{F,H} \frac{Hq}{J} \tau(d). \quad (61)$$

The zero frequency must remain excluded. Resolving exact conductors before this sum and applying [theorem 4.2](#) gives

$$\mathfrak{L}_{Y,S}(F_1, F_2) \ll_{\varepsilon} Q F_1 F_2 X^{\varepsilon} \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)}. \quad (62)$$

This is exactly the all-frequency argument of Lemmas 6.1–6.2 and Theorem 6.3 of TPC-22 [7]: replacing R by S changes only the ambient ranges $q \leq S^2$ and $F_i \leq S$. The Ramanujan-frequency sum in (61) cancels the Poisson normalization, while the two primitive Gauss factors and Cauchy–Schwarz contribute $F_1 F_2$.

Combining the two estimates for the same scalar cell proves the dynamic cell bound

$$\boxed{\mathfrak{L}_{Y,S}(F_1, F_2) \ll_{\varepsilon} Q X^{\varepsilon} \sqrt{\mathcal{K}_Q(F_1) \mathcal{K}_Q(F_2)} \min\{\sqrt{JY}, F_1 F_2\}.} \quad (63)$$

The omitted block $(F_1, F_2) = (1, 1)$ is the principal mean already treated in [theorem 3.2](#); one-sided principal coordinate cells remain in (63). Linearity and the pairwise triangle inequality give the formal reassembly

$$\sum_{q \asymp Y} |\mathcal{L}_{q,S}^{\text{disc}}| \leq \sum_{\substack{F_1, F_2 \\ (F_1, F_2) \neq (1, 1)}} \mathfrak{L}_{Y,S}(F_1, F_2) + O_A(X^{-A} \mathcal{Z}_1), \quad (64)$$

where the final term is the inherited polynomial Fourier-tail error.

Remark 4.3 (What was needed to cross R). Neither the character algebra nor the energy estimates require $L > 2S$. The inherited condition $L > 2R$ belongs to the row/source construction. For a target divisor $u > R$, the necessary primitivity is supplied by (15). The genuinely new analytic restriction is the Poisson-short condition (35).

5 Exponent ledger and a closed strip beyond R

Write

$$Q = X^{\beta+o(1)}, \quad J = X^{1-\beta+o(1)}, \quad S = X^{s+o(1)}, \quad (65)$$

and, on a denominator and conductor cell,

$$Y = X^{y+o(1)}, \quad F_i = X^{f_i+o(1)}. \quad (66)$$

The dynamic support gives

$$0 \leq y \leq 2s, \quad 0 \leq f_i \leq s. \quad (67)$$

Put

$$k_i = \max\{f_i, \beta - f_i\}; \quad (68)$$

then $\mathcal{K}_Q(F_i) = X^{k_i+o(1)}$.

The natural scale has exponent $1 + \beta$. The norm and direct-Gauss bounds in (63) have respective exponents

$$\beta + \frac{1 - \beta + y + k_1 + k_2}{2}, \quad \beta + f_1 + f_2 + \frac{k_1 + k_2}{2}.$$

Their saving exponents are therefore

$$\eta_A = \frac{\beta + 1 - y - k_1 - k_2}{2}, \quad \eta_B = 1 - f_1 - f_2 - \frac{k_1 + k_2}{2}. \quad (69)$$

Proposition 5.1 (Dynamic three-cell ledger). *Every cell in (67) satisfies*

$$\max\{\eta_A, \eta_B\} \geq \eta_S(\beta, s), \quad (70)$$

where

$$\eta_S(\beta, s) = \min \left\{ 1 - \frac{3\beta}{2}, \frac{\beta + 1 - 4s}{2}, \frac{3 - \beta - 6s}{4} \right\}. \quad (71)$$

Proof. Decreasing y improves η_A , so take $y = 2s$.

If $f_1, f_2 \leq \beta/2$, then $k_1 + k_2 = 2\beta - f_1 - f_2$, and

$$\eta_B = 1 - \beta - \frac{f_1 + f_2}{2} \geq 1 - \frac{3\beta}{2}.$$

If $f_1, f_2 \geq \beta/2$, then $k_1 + k_2 = f_1 + f_2 \leq 2s$, so

$$\eta_A \geq \frac{\beta + 1 - 4s}{2}.$$

In a mixed cell, write $x \leq \beta/2 \leq z$ and $d = z - x \geq 0$. Then $k_1 + k_2 = \beta + d$, whence

$$\eta_A \geq \frac{1}{2} - s - \frac{d}{2}. \quad (72)$$

Since $z \leq s$, direct substitution gives

$$\eta_B = 1 - 2z - \frac{\beta}{2} + \frac{d}{2} \geq 1 - 2s - \frac{\beta}{2} + \frac{d}{2}. \quad (73)$$

The maximum is at least the average of the two displayed lower bounds, namely $(3 - \beta - 6s)/4$. $\square$

Theorem 5.2 (Dynamic-cutoff closure). *Assume the primitive interface of section 2, including the row-pair-only off-diagonal mask, no-wrap condition, polynomial Fourier-tail mass, and the strict Poisson-short condition*

$$s < 1 - \beta \quad (74)$$

with a fixed margin. If

$$\eta_S(\beta, s) > 0 \quad (75)$$

with a fixed margin, then some $\eta > 0$ satisfies

$$\sum_{q \leq S^2} |\mathcal{L}_{q,S}^{\text{disc}}| \ll_{\epsilon, F, H} X Q X^{-\eta + \epsilon}. \quad (76)$$

Together with theorem 3.2, this closes the selected packet $u, v \leq S$, after the inherited dyadic and separated-packet reassembly.

Proof. Apply proposition 5.1 to every dyadic nonprincipal coordinate-conductor cell in (64), using (63). The fixed margin absorbs the X^ϵ , divisor-pair, and logarithmic conductor multiplicities. Condition (74) is precisely (35) at exponent level. The inherited no-wrap and tail-mass hypotheses license the norm route and discard the smooth Fourier tail. The principal cell is bounded by theorem 3.2 at $Q^2 X^\epsilon = (XQ/J)X^\epsilon$, which is also power-saving because $J = X^{1-\beta+o(1)}$. $\square$

There is a useful alternative on the short-period side. If the row source has the stronger margin $L > 2S$, the TPC-21 deterministic occupancy proof extends with S in place of R and closes the enlarged packet under

$$2s < 1 - \beta. \quad (77)$$

The extra condition $L > 2S$ is needed for that particular proof to replace $O(L/u + 1)$ by $O(L/u)$; it is not needed in theorem 5.2. We keep the hybrid theorem as the main cutoff-independent statement.

5.1 The polynomial collar

Return to $R = X^{1/2-\delta+o(1)}$ and put

$$S = RX^\rho, \quad s = \frac{1}{2} - \delta + \rho. \quad (78)$$

Corollary 5.3 (A closed large-divisor strip). *Suppose*

$$1 - 4\delta < \beta < \min \left\{ \frac{2}{3}, 6\delta \right\}, \quad \beta < \frac{1}{2} + \delta, \quad (79)$$

with fixed margins. Then [theorem 5.2](#) applies for every fixed $\rho > 0$ satisfying

$$\boxed{\rho < \min \left\{ \frac{1}{2} + \delta - \beta, \frac{\beta + 4\delta - 1}{4}, \frac{6\delta - \beta}{6} \right\}} \quad (80)$$

with a fixed margin. Hence the closed support grows from $[1, R]^2$ to $[1, RX^\rho]^2$.

Proof. The three quantities in (80) are respectively the strict Poisson-short margin and the positivity margins of the second and third entries in (71). The first entry of (71) is positive exactly when $\beta < 2/3$. $\square$

The conclusion contains all pairs in

$$[1, R] \times (R, S], \quad (R, S] \times [1, R], \quad (R, S]^2. \quad (81)$$

Thus it closes not only two mixed collars but also a genuine both-large square. This is a support statement about the residual coefficient; it is stronger than completing a single Fourier channel at fixed (u, v) .

Corollary 5.4 (Poisson-short frontier for $3/5 < \beta < 2/3$). *If $3/5 < \beta < 2/3$, then every fixed-power cutoff*

$$S \leq JX^{-\eta_0} \quad (82)$$

with $S > R$ and fixed $\eta_0 > 0$ satisfies all three conductor inequalities in (71) with a fixed margin. Hence the hybrid method reaches the strict Poisson-short boundary.

Proof. At the endpoint $s = 1 - \beta$, the second and third numerators in (71) both become $5\beta - 3$. They are positive for $\beta > 3/5$. Moving a fixed distance inside the endpoint supplies the required strict margins. $\square$

5.2 Uniform collar widths in the inherited profiles

For the published profile, let

$$\frac{11}{21} - 2\sigma < \beta \leq \frac{61}{84} - \sigma - \frac{\delta}{2}, \quad \delta = \frac{19}{126} - \frac{2\sigma}{3} + t, \quad (83)$$

where $0 < \sigma < 1/100$ and $t > 0$, subject to the inherited admissibility conditions $0 < \delta < 1/4$ and a nonempty displayed β -interval. Uniformly over its entire β -interval, any fixed

$$0 < \rho < \min \left\{ \frac{3t}{2}, \frac{2}{63} - \frac{7\sigma}{6} + t, \frac{8}{189} - \frac{5\sigma}{9} + \frac{13t}{12} \right\} \quad (84)$$

is allowed.

For the version-locked second profile,

$$\frac{9}{17} - \frac{\sigma}{2} < \beta \leq \frac{49}{68} - \sigma - \frac{\delta}{2}, \quad \delta = \frac{5}{34} - \frac{2\sigma}{3} + t, \quad (85)$$

with $0 < \sigma < 1/1000$, subject to $0 < \delta < 1/4$ and a nonempty displayed β -interval, a uniform choice is

$$0 < \rho < \min \left\{ \frac{3t}{2}, \frac{1}{34} - \frac{19\sigma}{24} + t, \frac{2}{51} - \frac{5\sigma}{9} + \frac{13t}{12} \right\}. \quad (86)$$

The first entry in both ledgers tends to zero as $t \downarrow 0$: an endpoint equality supplies no fixed-power collar.

6 Complementary divisors and affine parity fibers

The dynamic theorem stops before the Poisson-short or conductor margin is exhausted. We now give an exact normal form for what remains. No estimate in this section assumes that a target is squarefree.

6.1 Hyperbola reflection

For $n \geq 1$ and $Y \geq 1$, put

$$\mathcal{T}_Y(n) = \sum_{\substack{u|n \\ u > Y}} a(u), \quad a(u) = -\mu(u) \log u. \quad (87)$$

Proposition 6.1 (Exact complement and Liouville cocycle). *For every $n, Y \geq 1$,*

$$\mathcal{T}_Y(n) = \sum_{\substack{r|n \\ r < n/Y}} a(n/r) \quad (88)$$

$$= -\lambda(n) \sum_{\substack{r|n \\ r < n/Y}} \lambda(r) \mu^2(n/r) \log(n/r). \quad (89)$$

Here λ is the Liouville function. In particular, the sign on the reflected quotient is $\mu(n/r)$, not $\mu(r)$.

Proof. The involution $u \mapsto r = n/u$ bijects divisors satisfying $u > Y$ with those satisfying $r < n/Y$, proving (88). Since $\mu(k) = \lambda(k) \mu^2(k)$ and λ is completely multiplicative,

$$\mu(n/r) = \lambda(n/r) \mu^2(n/r) = \lambda(n) \lambda(r) \mu^2(n/r).$$

Insert this identity into $a(n/r)$. □

If $n > Y^2$, the tail has the exact hyperbola split

$$\mathcal{T}_Y(n) = \underbrace{\sum_{\substack{r|n \\ r \leq Y}} a(n/r)}_{\mathcal{C}_Y(n)} + \underbrace{\sum_{\substack{r|n \\ Y < r < n/Y}} a(n/r)}_{\mathcal{B}_Y(n)}, \quad (90)$$

with the evident adjustment when a boundary divisor occurs. The first piece has a short complementary divisor. In the second, both factors of $n = r(n/r)$ exceed Y ; it is a genuine Type-II hyperbola band. Pairing divisors about $\sqrt{n}$ gives

$$\mathcal{B}_Y(n) = \sum_{\substack{d|n \\ Y < d < \sqrt{n}}} \{a(d) + a(n/d)\} + \mathbf{1}_{n=z^2, z > Y} a(z). \quad (91)$$

On a squarefree target, the paired coefficient is

$$a(d) + a(n/d) = -\lambda(d)\{\log d + \lambda(n)\log(n/d)\}. \quad (92)$$

Thus even and odd factor parity are different eigensectors of divisor reflection. This is an identity, not a bound for either sector.

For the TPC target $n \asymp X$ and $Y = R = X^{1/2-\delta}$, the balanced factor exponents lie between $1/2 - \delta$ and $1/2 + \delta$. The dynamic theorem removes a strict lower part of this geometry before the quotient-Möbius normal form is needed.

6.2 Two complement fibers

Fix distinct positive primitive rows m_1, m_2 , meaning $(m_1 m_2, h) = 1$, and write

$$n_i(j) = m_i j + h \quad (i = 1, 2). \quad (93)$$

Let $r, s \geq 1$. Since $(m_1 m_2, h) = 1$, any solution of $r \mid n_1(j)$, $s \mid n_2(j)$ necessarily satisfies

$$(r, m_1) = (s, m_2) = 1. \quad (94)$$

We restrict to such pairs and put

$$d = (r, s), \quad q = [r, s] = \frac{rs}{d}. \quad (95)$$

Theorem 6.2 (Exact affine complement fiber). *The two congruences*

$$r \mid m_1 j + h, \quad s \mid m_2 j + h \quad (96)$$

are jointly soluble if and only if

$$d \mid h(m_1 - m_2). \quad (97)$$

When they are soluble, they select a unique class $j_0 \pmod{q}$. Writing $j = j_0 + qt$, define

$$U_{r,s}(t) = \frac{m_1(j_0 + qt) + h}{r} = A_1 t + B_1, \quad (98)$$

$$V_{r,s}(t) = \frac{m_2(j_0 + qt) + h}{s} = A_2 t + B_2. \quad (99)$$

These are integer affine forms with

$$\boxed{A_1 B_2 - A_2 B_1 = \frac{h(m_1 - m_2)}{d} \neq 0.} \quad (100)$$

Consequently, for any finitely supported orbit weight W ,

$$\begin{aligned} & \sum_j W(j) \mathcal{T}_S(n_1(j)) \mathcal{T}_S(n_2(j)) \\ &= \sum_{\substack{r,s \geq 1 \\ (r,m_1)=(s,m_2)=1 \\ (r,s) \mid h(m_1-m_2)}} \sum_{t \in \mathbb{Z}} W(j_0 + qt) a(U_{r,s}(t)) a(V_{r,s}(t)) \mathbf{1}_{U_{r,s}(t) > S} \mathbf{1}_{V_{r,s}(t) > S}. \end{aligned} \quad (101)$$

All terms with nonpositive targets or outside the finite horizon are understood to be absent.

Proof. Each congruence in (96) selects one class modulo its divisor. Their reductions modulo d agree exactly when $-h\overline{m_1} \equiv -h\overline{m_2} \pmod{d}$. Since $(m_1 m_2, d) = 1$, multiplication by $m_1 m_2$ shows that this is equivalent to (97). The Chinese remainder theorem then gives one class modulo q .

The quantities in (98)–(99) are integers, with

$$A_1 = \frac{m_1 q}{r}, \quad B_1 = \frac{m_1 j_0 + h}{r}, \quad A_2 = \frac{m_2 q}{s}, \quad B_2 = \frac{m_2 j_0 + h}{s}.$$

Direct calculation gives

$$A_1 B_2 - A_2 B_1 = \frac{q h (m_1 - m_2)}{r s} = \frac{h (m_1 - m_2)}{d}.$$

It is nonzero because the generic mask has $m_1 \neq m_2$. Finally, expand both tails using (88), group by (r, s) , and parameterize the common progression. The inequalities on U, V are exactly the original requirements that the quotient divisors exceed S . $\square$

Because $a(U)a(V) = \mu(U)\mu(V) \log U \log V$, (101) is an exact family of two-point affine Möbius correlations. The determinant content $d = (r, s)$ divides $h(m_1 - m_2)$. On the primitive orbit $(j, H) = 1$, it is coprime to H and therefore divides the row difference. In a mixed large–short leg there is also the original divisor $\gcd(U, v) \mid m_1 - m_2$; the two contents can overlap when the target has square factors. They must not be declared coprime.

Remark 6.3 (The $r = s = 1$ parity core). The first complement fiber is always compatible and has

$$U_{1,1}(j) = m_1 j + h, \quad V_{1,1}(j) = m_2 j + h. \quad (102)$$

Its coefficient is

$$\mu(m_1 j + h) \mu(m_2 j + h) \log(m_1 j + h) \log(m_2 j + h). \quad (103)$$

On prime targets, each individual factor $-\mu(m_i j + h) \log(m_i j + h)$ is the positive prime weight. Thus a generic assertion that this fiber is negligible by “Möbius randomness” would be a parity-level overclaim; the precise averaging and uniformity are indispensable.

7 A logarithmic bridge and an absolute-value no-go

The affine normal form permits one rigorous step through the parity gate, but only in a weaker averaging regime and without the uniformity required by the TPC packet.

7.1 Every fixed fiber cancels logarithmically

Tao proved the logarithmically averaged two-point Elliott conjecture [2]. In the special case needed here, if

$$L_1(t) = A_1 t + B_1, \quad L_2(t) = A_2 t + B_2, \quad A_1 B_2 - A_2 B_1 \neq 0, \quad (104)$$

are fixed and positive for large t , then

$$\sum_{t \leq T} \frac{\mu(L_1(t)) \mu(L_2(t))}{t} = o(\log T). \quad (105)$$

This is Corollary 5 of Tao [2], applied with $g_1 = g_2 = \mu$ and $\omega(T) = T$. The Möbius function satisfies the corrected strong nonpretentiousness condition used there; Tao deduces this from the classical zero-free region. A fixed initial shift may be made first if one of the constants B_i is negative, so both forms lie in the positive integers throughout the sum.

Proposition 7.1 (Log-weighted cancellation on a fixed fiber). *Under (104),*

$$\boxed{\sum_{t \leq T} \frac{a(L_1(t))a(L_2(t))}{t} = o((\log T)^3).} \quad (106)$$

In particular, every fixed fiber in theorem 6.2 has logarithmically averaged cancellation.

Proof. Set

$$A(x) = \sum_{t \leq x} \frac{\mu(L_1(t))\mu(L_2(t))}{t} = o(\log x)$$

by (105). The weight $w(t) = \log L_1(t) \log L_2(t)$ satisfies

$$w(t) \ll (\log(2t))^2, \quad w'(t) \ll \frac{\log(2t)}{t}$$

away from a fixed initial interval. Partial summation gives

$$A(T)w(T) - \int_2^T A(t)w'(t) dt = o((\log T)^3).$$

The two minus signs in $a(L_1)a(L_2)$ cancel. □

For a fixed complement cutoff Z , define

$$\mathcal{T}_m^{[Z]}(j) = \sum_{\substack{r|mj+h \\ r \leq Z}} a((mj+h)/r). \quad (107)$$

Corollary 7.2 (Finite complement reassembly). *Fix $h \neq 0$, distinct positive rows m_1, m_2 coprime to h , and $Z \geq 1$. Then*

$$\boxed{\sum_{j \leq J} \frac{\mathcal{T}_{m_1}^{[Z]}(j)\mathcal{T}_{m_2}^{[Z]}(j)}{j} = o_{m_1, m_2, h, Z}((\log J)^3).} \quad (108)$$

Proof. Expand the two finite divisor sums. For each fixed $r, s \leq Z$, the two divisibility conditions are either incompatible or select one class $j = j_0 + [r, s]t$. In the latter case, theorem 6.2 gives two affine quotient forms with nonzero determinant. Also

$$\frac{1}{j_0 + [r, s]t} = \frac{1}{[r, s]t} + O_{r,s}(t^{-2}).$$

Apply proposition 7.1; the t^{-2} error converges after the logarithmic weights. There are only finitely many pairs. □

The fixedness can be relaxed only ineffectively by a diagonal argument.

Corollary 7.3 (An arbitrarily slow moving window). *For fixed $h \neq 0$, there exists a nondecreasing function $K(J) \rightarrow \infty$ such that*

$$\max_{\substack{1 \leq m_1, m_2, Z \leq K(J) \\ m_1 \neq m_2 \\ (m_1 m_2, h) = 1}} \frac{1}{(\log J)^3} \left| \sum_{j \leq J} \frac{\mathcal{T}_{m_1}^{[Z]}(j)\mathcal{T}_{m_2}^{[Z]}(j)}{j} \right| \rightarrow 0. \quad (109)$$

Proof. For each integer K , corollary 7.2 holds uniformly over the finite set of triples $m_1, m_2, Z \leq K$, with the maximum over an empty set interpreted as zero. Choose strictly increasing thresholds $J_K \geq K$ beyond which the normalized maximum is at most $1/K$, and set $K(J) = \max\{K : J_K \leq J\}$, with $K(J) = 1$ below J_1 . □

This is a genuine moving-parameter consequence, but it supplies no effective rate at all. It is many orders of magnitude weaker than the polynomial scales $m_i \asymp Q = X^\beta$ and $r, s \ll X/S$ in (101).

7.2 Why absolute values cannot replace the correlation

We next show that the unit-complement fiber is large after signs are removed. This is a componentwise obstruction, not a lower bound for the complete signed residual packet.

For fixed primitive rows, let ν_p be the number of solutions $j \pmod{p^2}$ of

$$(m_1j + h)(m_2j + h) \equiv 0 \pmod{p^2}. \quad (110)$$

Each form has at most one root modulo p^2 , and it has none if $p \mid m_i$, because then $p \nmid h$. Hence $0 \leq \nu_p \leq 2$, and

$$\mathfrak{S}_{\text{sf}}(m_1, m_2; h) = \prod_p \left(1 - \frac{\nu_p}{p^2}\right) > 0. \quad (111)$$

Lemma 7.4 (Simultaneous squarefree density). *For fixed $m_1 \neq m_2$ with $(m_1m_2, h) = 1$,*

$$\sum_{j \leq J} \mu^2(m_1j + h) \mu^2(m_2j + h) = \mathfrak{S}_{\text{sf}}(m_1, m_2; h)J + o(J). \quad (112)$$

Proof. For a fixed prime bound P , the Chinese remainder theorem gives the exact density $\prod_{p \leq P} (1 - \nu_p/p^2)$ of classes avoiding all square divisors p^2 with $p \leq P$, up to $O_P(1/J)$. The number of $j \leq J$ for which one of the two forms is divisible by p^2 for some $p > P$ is

$$\ll \sum_{P < p \ll \sqrt{J}} \left(\frac{J}{p^2} + 1\right) = O(J/P) + O(\sqrt{J}/\log J),$$

where the implied constants may depend on the fixed rows and h . Let $J \rightarrow \infty$ and then $P \rightarrow \infty$. Absolute convergence and positivity of (111) follow from $\nu_p \leq 2$. $\square$

Proposition 7.5 (Absolute-mass no-go). *For the unit-complement coefficient in (103),*

$$\boxed{\sum_{J/2 < j \leq J} |\mu(m_1j + h) \mu(m_2j + h)| \log(m_1j + h) \log(m_2j + h) \gg_{m_1, m_2, h} J(\log J)^2.} \quad (113)$$

Therefore a fiberwise triangle inequality or sign-blind estimate cannot make this fixed, unweighted component power-saving.

Proof. By lemma 7.4, a positive proportion of the integers in $(J/2, J]$ make both forms squarefree. On that interval both logarithms are $\asymp \log J$, and $|\mu(n)| = \mu^2(n)$. $\square$

There is no contradiction between propositions 7.1 and 7.5: the first retains the product of the two Möbius signs and uses logarithmic averaging; the second deliberately removes exactly that cancellation.

8 The remaining uniform gate and a stepwise roadmap

The dynamic theorem and complement theorem meet at a precise boundary. Inside $u, v \leq S$, exact-conductor averaging supplies a fixed-power saving. Outside, divisor reflection exposes affine Möbius correlations. This section records the order of operations needed to continue and the conclusions that are not available.

8.1 A quantitative form of the next gate

For a compatible complement pair, define the sharp fiber sum

$$\begin{aligned} \mathcal{M}_{m_1, m_2; r, s}(W; S) &:= \sum_t W \left(\frac{j_0 + [r, s]t}{J} \right) \mu(U_{r, s}(t)) \mu(V_{r, s}(t)) \\ &\quad \times \log U_{r, s}(t) \log V_{r, s}(t) \mathbf{1}_{U_{r, s}(t) > S} \mathbf{1}_{V_{r, s}(t) > S}. \end{aligned} \quad (114)$$

The smooth weight W ranges over a fixed family with uniformly bounded support and derivative seminorms. The forms and their nonzero determinant are given by [theorem 6.2](#). The fiber has length

$$\asymp 1 + \frac{J}{[r, s]}. \quad (115)$$

Tao's theorem controls (114) when every coefficient is fixed, the average is cumulative logarithmic, and the limit is taken before the coefficients grow. The TPC packet instead requires all of the following simultaneously:

- (i) $m_i \asymp Q = X^{\beta+o(1)}$ and complement divisors as large as X/S ;
- (ii) a prescribed smooth dyadic window of length J , not merely a cumulative dt/t average;
- (iii) a rate strong enough to sum the row pairs, complement pairs, determinant-content refinements, and separated smooth packets;
- (iv) a separate treatment of the sparse range $[r, s] \gtrsim J$, where a fiber has at most $O(1)$ orbit points and no one-fiber asymptotic can create cancellation.

The next useful theorem would therefore be a *uniform affine Möbius dispersion estimate*, not a pointwise appeal to randomness. One possible sufficient interface is the following.

Conditional input 8.1 (Dyadic affine-fiber dispersion). After inserting the actual row weights and generic row-pair mask, every dyadic block of compatible complement moduli with $[r, s] \leq JX^{-\kappa}$ has total contribution from (114) at most

$$XQ(\log X)^{-B} \quad (116)$$

for arbitrarily prescribed $B > 0$, uniformly over the separated weights. The complementary sparse blocks admit the same total bound after using their exact determinant and incidence geometry rather than a one-fiber asymptotic.

This is deliberately an input, not a theorem of the present paper. It keeps the two Möbius signs together and specifies the growing parameters, sharp averaging, coefficient summation, and sparse regime that a usable result must cover.

8.2 Why several shortcuts fail

- (1) *Taking absolute values fiber by fiber.* [Proposition 7.5](#) gives natural-scale mass already on $r = s = 1$. The two Möbius signs are the mechanism, not a nuisance that can be discarded.
- (2) *Replacing the reflected sign by $\mu(r)$.* The exact coefficient is $\mu(n/r)$; see (88). The Liouville rewriting introduces the global target phase $\lambda(n)$ and a squarefree quotient cutoff. It does not turn the coefficient into an ordinary short-divisor Möbius sum.
- (3) *Assuming the two determinant contents are coprime.* In a mixed large-short term, both the complement gcd and the quotient-short-divisor gcd divide $m_1 - m_2$, but they may overlap when the target has a square factor. Declaring them coprime silently deletes valid nonsquarefree targets.

- (4) *Applying TPC-22 with an arbitrarily large cutoff.* The dynamic conductor ledger loses its margin at the explicit faces in (71), and TPC-20's Fourier-channel reduction loses the strict Poisson-short condition at $S \asymp J$. Beyond those faces, additional cancellation between joint and single spectra or a new quotient-variable argument is required.
- (5) *Using logarithmic Elliott as a black box for the full packet.* Corollary 7.3 is intentionally ineffective. It supplies no bound uniform over polynomially many forms, and logarithmic averaging cannot simply be removed by partial summation.

8.3 A staged continuation

The exact reductions suggest the following sequence of independent tasks.

- Stage 1: Push the dynamic cutoff to the largest certified S on each row slice, using theorem 5.2. This has been completed here for the full square $u, v \leq S$.
- Stage 2: Split the remaining both-ultra terms by the complement modulus $q = [r, s]$, and the mixed terms by their complementary-short modulus. Preserve the row mask, the exact affine determinant $h(m_1 - m_2)/(r, s)$, and the corresponding mixed gcd content.
- Stage 3: In $q \ll J$, seek a quantitative version of two-point Möbius cancellation that is uniform in the polynomial coefficients generated by the complement fibers. A logarithmically averaged or averaged-row theorem would already be publishable if its precise range exceeds the current slow diagonal window.
- Stage 4: In $q \gtrsim J$, use sparse incidence geometry, the equation

$$m_2 U r - m_1 V s = h(m_2 - m_1), \tag{117}$$

and the generic determinant/gcd pruning. A theorem here need not solve an individual two-point Chowla problem because each fiber is short; it must instead save after summing the family.

- Stage 5: Recombine the long and sparse complement blocks with the mixed $Z^{\leq S} Z^{> S}$ terms. Only after this scalar recombination may one restore all separated packets and the earlier TPC-18 endpoint pieces.
- Stage 6: A full residual-dispersion estimate would still be an analytic upper/cancellation theorem. Converting it into a positive Hardy–Littlewood main term or an infinite twin-prime conclusion is a separate parity-sensitive step.

Each stage has a useful fallback. Failure of a uniform estimate can be recorded as a sharp exponent boundary or a componentwise no-go; success in a logarithmic, averaged-row, or shorter-parameter regime is still a rigorous intermediate theorem.

8.4 Status table

Object	Status after this paper
Ramanujan mean for $b_R(u)\mathbf{1}_{u \leq S}$	Closed absolutely by theorem 3.2 .
Centered connected packet $u, v \leq S$	Closed with fixed-power saving in theorem 5.2 and corollary 5.3 .
Mixed and both-large pieces newly included in (81)	Closed; these are genuine coefficients beyond R .
Remaining terms with $u > S$ or $v > S$	Exact complement-fiber form proved; polynomially uniform estimate open.
Every fixed complement fiber and fixed finite complement truncation	Closed only in cumulative logarithmic average by proposition 7.1 and corollary 7.2 .
Full TPC-18 residual and all endpoint reassembly	Not proved.
Hardy–Littlewood asymptotic, twin-prime lower bound, or a general parity breakthrough	Not claimed and not implied.

9 Exact certificate and conclusion

The analytic proofs above are symbolic. We include a deterministic finite certificate to audit the identities and rational exponent ledger most vulnerable to normalization, boundary, or gcd errors. It does not test the logarithmically averaged theorem, estimate primes, or replace any asymptotic argument.

9.1 Certificate

The script

`experiments/tpc23_certificate.py`

uses only the Python standard library. It represents $-\mu(n) \log n$ exactly as a vector on the formal basis $\{\log p : p \text{ prime}\}$, and uses `fractions.Fraction` for every exponent calculation. It checks:

- (1) complement reflection (88) for all $2 \leq n \leq 400$ and $1 \leq Y \leq 24$, including nonsquarefree targets;
- (2) the Liouville cocycle (89) and balanced hyperbola split on the same models;
- (3) the general CRT compatibility $(r, s) \mid h(m_1 - m_2)$ and affine determinant (100) over five primitive row-shift samples and a finite modulus family that includes complements sharing primes with h ;
- (4) a hard-failure variant which omits the factor (r, s) from the determinant and is required to disagree;
- (5) the dynamic three-cell inequality on an exact rational grid;
- (6) both inherited profile collar identities (84) and (86).

Independent ordinary and `python -O` runs perform 56,714 exact checks and produce byte-identical output. The hashes are

JSON SHA-256	62e982d5da46d68fcf698f498a2d8fa66cb54ca9422d470f8516bc18f2aa08ab
source SHA-256	51640ec4cbc5c82d792ce40a170ec919ffa2593fade6f92eaffca25d612429be
certificate digest	c5054fcce65ccbda59dba99c519f5fd9fbe147384f63f6b5d0e17265876be97

The JSON flags for asymptotic evidence, full residual closure, Hardy–Littlewood, twin primes, and a general parity breakthrough are all false.

9.2 Conclusion

The large-divisor support left after TPC-22 is not an indivisible black box. The exact-conductor method is stable under a dynamic residual cutoff, and its new exponent ledger proves that every strict interior point of the old hybrid closure region contains a nonzero collar beyond R . The enlarged square $u, v \leq S$ includes two mixed pieces and a both-large piece. This is a genuine fixed-power analytic advance, not a numerical observation or a renamed short-short theorem.

At the next boundary, divisor reflection changes the problem. It shortens the complementary modulus but transfers the coefficient to the quotient $\mu((mj + h)/r)$. On two rows, the quotient variables are nonproportional affine forms with exact determinant $h(m_1 - m_2)/(r, s)$. This identifies a real two-point parity cocycle. Known logarithmic Elliott theory cancels each fixed fiber and every fixed finite complement family; a diagonal argument even permits an ineffective slowly growing family. The absolute-mass theorem shows why the signs cannot be removed.

What remains is now narrower and more explicit: obtain a rate uniform over polynomially growing affine coefficients and complement fibers in the long range, and a summed incidence estimate in the sparse range. Either advance would move the full residual program forward. Neither is supplied here, and neither may be replaced by an assertion of random Möbius signs. Full residual reassembly, a positive prime-pair main term, and the twin-prime conjecture remain separate open stages.

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